



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Question Bank

Programme:	B.E	Department:	CSE
Academic Year:	2024-2025	Year/Sem/Sec:	III/V/I&II
Course Code:	231CSV68	Course Name:	SQL FOR DATA SCIENCE
Course Tutor:	R.KESAVAN,AP/CSE		

UNIT I – COMPUTATIONAL THINKING AND PROBLEM SOLVING

Q. No.	PART –A	Marks	Bloom's	COs
1.	Which of the following is a type of SQL command used to retrieve data? a) INSERT b) DELETE c) SELECT d) UPDATE	1	K1	CO1
2.	Which SQL keyword is used to filter records based on a condition? a) FILTER b) SORT c) WHERE d) GROUP	1	K1	CO1
3.	What does RDBMS stand for? a) Rational Database Management System b) Remote Database Management System c) Relational Database Management System d) Reliable Database Management System	1	K1	CO1
4.	Which SQL keyword is used to insert new data into a table? a) INSERT b) ADD c) INCLUDE d) PUT	1	K1	CO1
5.	Which clause is used to sort data in ascending or descending order? a) GROUP BY b) SORT c) ORDER BY d) ARRANGE	1	K1	CO1
6.	Which of the following is a valid SQL data type? a) NUMERIC b) VARCHAR c) TEXTUAL d) CHARACTERS	1	K1	CO1
7.	Which constraint ensures a column value is unique and not null? a) FOREIGN KEY b) PRIMARY KEY c) DEFAULT d) CHECK	1	K1	CO1
8.	Which keyword is used to change existing data in a table? a) ALTER b) UPDATE c) MODIFY d) CHANGE	1	K1	CO1
9.	Which command is used to remove data from a table without deleting the table? a) DROP b) DELETE c) REMOVE d) ERASE	1	K1	CO1
10.	What data type is used to store whole numbers in SQL? a) DECIMAL b) INT c) FLOAT d) DOUBLE	1	K1	CO1

11.	Which SQL clause is used to rename a column in the output? a) AS b) RENAME c) REPLACE d) NAME	1	K1	CO1
12.	Which constraint restricts values in a column to meet a specific condition? a) NOT NULL b) DEFAULT c) CHECK d) UNIQUE	1	K1	CO1
13.	The command to create a new table in SQL is: a) ADD TABLE b) MAKE TABLE c) CREATE TABLE d) INSERT TABLE	1	K1	CO1
14.	To avoid duplicate entries in a column, which constraint is used? a) UNIQUE b) NOT NULL c) DEFAULT d) INDEX	1	K1	CO1
15.	What keyword is used to define a default value for a column? a) SET b) DEFAULT c) VALUE d) FIXED	1	K1	CO1
16.	Which of the following is not a valid SQL command? a) SELECT b) GET c) INSERT d) DELETE	1	K1	CO1
17.	In SQL, the ‘%’ symbol is used with which operator? a) LIKE b) IN c) BETWEEN d) AND	1	K1	CO1
18.	Which keyword is used to remove all rows from a table quickly? a) TRUNCATE b) DROP c) DELETE d) CLEAR	1	K1	CO1
19.	Which of the following constraints enforces referential integrity? a) FOREIGN KEY b) PRIMARY KEY c) UNIQUE d) CHECK	1	K1	CO1
20.	Which keyword is used to sort data in descending order? a) ASC b) DESC c) DOWN d) LOW	1	K1	CO1
PART B				
1.	What is the difference between DBMS and RDBMS?	2	K1	CO1
2.	Define a Primary Key with an example.	2	K1	CO1
3.	What is the use of the SELECT statement in SQL?	2	K1	CO1
4.	Write the syntax of the INSERT statement with an example.	2	K1	CO1
5.	Differentiate between DELETE and TRUNCATE commands.	2	K1	CO1
6.	What are SQL constraints? Name any four.	2	K1	CO1
7.	Explain the use of the WHERE clause in SQL.	2	K1	CO1
8.	Write a query to display all employees in descending order of their salary.	2	K1	CO1
9.	What is the difference between CHAR and VARCHAR data types?	2	K1	CO1
10.	Write a query to update the name of a student with id = 101.	2	K1	CO1
11.	Define a Foreign Key with an example.	2	K1	CO1
12.	Write the SQL command to create a table named "products" with columns: id (int), name (varchar), price (float).	2	K1	CO1
13.	What is the purpose of the ORDER BY clause? Give an example.	2	K1	CO1
14.	Write a query to delete all students with name = 'Rahul'.	2	K1	CO1
15.	How do you enforce a column to always have a value in SQL?	2	K1	CO1
PART C				

1.	Analyze and write SQL queries to demonstrate the use of SELECT, INSERT, UPDATE, and DELETE commands on a student database. Justify the use of each command with examples.	14	K1	CO1
2.	Evaluate different types of constraints in SQL. Illustrate how each constraint affects data integrity with SQL code examples.	14	K1	CO1
3.	Compare and contrast the use of WHERE and ORDER BY clauses with complex queries. Highlight how they impact result sets.	7	K1	CO1
4.	Examine the various SQL data types used in database design. Provide real-world examples for selecting suitable data types.	14	K1	CO1
5.	Formulate complex SQL queries using SELECT, WHERE, and ORDER BY clauses to retrieve meaningful insights from a company's employee database.	14	K1	CO1
6.	Analyze the various categories of SQL commands and evaluate their roles in managing both the structure and data of a relational database. Support your analysis with appropriate SQL examples.	14	K1	CO1
7.	Critically examine the different types of constraints in SQL. How do these constraints collectively enforce data integrity and consistency within relational databases? Illustrate your answer with complex use-case examples.	14	K1	CO1

UNIT II:DATA TYPES, EXPRESSIONS, STATEMENTS				
1.	Which JOIN returns only the matching rows from both tables? a) LEFT JOIN b) RIGHT JOIN c) INNER JOIN d) FULL JOIN	1	K1	CO2
2.	Which JOIN returns all rows from the left table and matched rows from the right? a) FULL JOIN b) LEFT JOIN c) INNER JOIN d) CROSS JOIN	1	K1	CO2
3.	What does a FULL JOIN return? a) Only matched rows b) Only unmatched rows c) All matched and unmatched rows from both tables d) Rows from right table only	1	K1	CO2
4.	What keyword is used to combine the result of two SELECT queries with duplicate rows removed? a) UNION b) INTERSECT c) EXCEPT d) JOIN	1	K1	CO2
5.	Which operation returns rows present in both queries? a) UNION b) INTERSECT c) EXCEPT d) FULL JOIN	1	K1	CO2
6.	Which operation returns rows from the first query that are not in the second? a) UNION b) INTERSECT c) EXCEPT d) CROSS JOIN	1	K1	CO2
7.	A subquery inside the main query is called a: a) Self join b) Nested query c) Inline join d) External query	1	K1	CO2
8.	What clause is commonly used with subqueries? a) ADD b) MERGE c) WHERE d) SPLIT	1	K1	CO2
9.	Which SQL clause is used for conditional logic? a) LOOP b) IF-ELSE c) CASE d) CHECK	1	K1	CO2
10.	What keyword is used in SQL to check multiple conditions in a SELECT? a) CASE b) LOOP c) TEST d) WHILE	1	K1	CO2

11	What is the default join type in SQL when only JOIN is written? a) LEFT JOIN b) INNER JOIN c) RIGHT JOIN d) FULL JOIN	1	K1	CO2
12	Which keyword is not allowed in a subquery? a) SELECT b) FROM c) ORDER BY d) WHERE	1	K1	CO2
13	How many tables does an INNER JOIN need at minimum? a) 1 b) 2 c) 3 d) 0	1	K1	CO2
14	What will the UNION ALL keyword do that UNION does not? a) Sort results b) Include duplicates c) Remove NULLs d) Join tables	1	K1	CO2
15	Which clause is used in SQL to filter rows based on a condition in a subquery? a) HAVING b) WHERE c) ORDER BY d) GROUP BY	1	K1	CO2
16	What is returned by a RIGHT JOIN? a) Only matched rows b) All rows from the right and matched rows from the left c) All rows from the left d) Duplicate rows only	1	K1	CO2
17	Which of the following is a set operator in SQL? a) INTERSECT b) CASE c) COUNT d) GROUP	1	K1	CO2
18	CASE statement can be used in which clause? a) FROM b) GROUP BY c) SELECT d) CREATE	1	K1	CO2
19	Subqueries must always be enclosed in: a) Square brackets b) Parentheses c) Curly braces d) Quotes	1	K1	CO2
20	Which function is used to test conditions in SQL apart from CASE? a) EVALUATE b) SWITCH c) IF() d) TEST()	1	K1	CO2
PART B				
1.	What is the difference between INNER JOIN and LEFT JOIN?	2	K1	CO2
2.	Write a query using a subquery to find employees with salary greater than the average salary.	2	K1	CO2
3.	What does the UNION operator do in SQL?	2	K1	CO2
4.	Explain the use of the CASE statement in SQL.	2	K1	CO2
5.	Write a query using RIGHT JOIN to display departments and their employees.	2	K1	CO2
6.	What is a correlated subquery?	2	K1	CO2
7.	How is EXCEPT different from INTERSECT in SQL?	2	K1	CO2
8.	Give an example using UNION ALL.	2	K1	CO2
9.	Write a query that uses CASE to classify students based on marks.	2	K1	CO2
10.	Explain the concept of FULL JOIN with example.	2	K1	CO2
11.	What is a nested SELECT? Give an example.	2	K1	CO2
12.	Can we use ORDER BY in subqueries? Why or why not?	2	K1	CO2
13.	How does IF() function work in MySQL?	2	K1	CO2
14.	Differentiate between JOIN and UNION.	2	K1	CO2
15.	Write a query to display common names in two tables using INTERSECT.	2	K1	CO2
PART C				
1.	Explain different types of JOIN operations in SQL with syntax and examples.	14	K2	CO2
2.	Describe the sub query and different its types.	14	K2	CO2
3.	Explain the use of UNION, INTERSECT, and EXCEPT operators in SQL. Provide syntax and examples.	14	K2	CO2
4.	Write detailed SQL queries using CASE and IF conditions. Explain use cases and	7	K1	CO2

	benefits.			
5.	Compare and contrast JOINS and set operations (UNION, INTERSECT, EXCEPT). Write examples to show the differences.	7	K2	CO2

UNIT III CONTROL FLOW AND FUNCTIONS				
1.	Which function returns the total number of rows in a table? a) SUM b) AVG c) COUNT d) TOTAL	1	K1	CO4
2.	Which SQL function is used to calculate the average value of a column? a)MIN b)SUM c)AVG d) MAX	1	K1	CO4
3.	Which clause is used to group rows sharing a common field? a) MIN b) MAX c) SUM d) COUNT	1	K1	CO4
4.	What does the HAVING clause do? a) Filters rows before grouping b) Sorts the result set c) Filters groups after grouping d) Deletes NULL rows	1	K1	CO4
5.	What function returns the total sum of a column? a) COUNT b) AVG c) SUM d) MAX	1	K2	CO4
6.	Which clause is used to group rows sharing a common field? a) ORDER BY b) GROUP BY c) HAVING d) DISTINCT	1	K1	CO4
7.	Which function returns the smallest value in a column? a) MIN b) AVG c) MAX d) SUM	1	K1	CO4
8.	Which clause must be used with GROUP BY to filter groups based on a condition? a) WHERE b) HAVING c) ORDER BY d) LIMIT	1	K1	CO4
9.	Which window function gives the same rank to equal values but skips the next ranks? a) ROW_NUMBER b) RANK c) COUNT d) AVG	1	K2	CO4
10	Which window function does not skip ranks for duplicates? a) RANK b) ROW_NUMBER c) DENSE_RANK d) SUM	1	K1	CO4
11	Which window function assigns a unique number to each row regardless of values? a) COUNT b) ROW_NUMBER c) RANK d) MAX	1	K1	CO4
12	Which function returns the current date in SQL? a) GETDATE b) CURRENT_DATE c) DATE_NOW d) TODAY	1	K1	CO4
13	What is the result of COUNT(*)? a) Count of non-NULL values b) Count of all rows including NULLs c) Count of zero d) Error	1	K1	CO4

14	Which SQL function gets the current time? a) CURRENT_TIME b) TIME_NOW c) NOW_DATE d) DATE_TIME	1	K2	C04
15	Which clause comes after GROUP BY in a SQL query? a) SELECT b) FROM c) HAVING d) WHERE	1	K2	C04
16	What does NOW() function return? a) Only current date b) Current date and time c) Date from last week d) Yesterday's date	1	K1	C04
17	What will AVG(salary) return? a) Sum of salary b) Average salary c) Minimum salary d) Row count	1	K1	C04
18	Which function helps retrieve records within a ranking in SQL? a) RANK() b) MIN() c) IF() d) ORDER BY	1	K2	C04
19	Which clause is not required for using aggregate functions? a) HAVING b) SELECT c) FROM d) GROUP BY	1	K2	C04
20	Which function is used to calculate the difference between two dates in SQL? a) TIMESTAMP() b) DATEDIFF() c) DATE_ADD() d) TIMEZONE()	1	K1	C04
PART B				
1.	What is the use of COUNT() function in SQL?	2	K1	C03
2.	Write a query to get the total sales from the orders table.	2	K1	C03
3.	Explain the purpose of the GROUP BY clause with an example.	2	K1	C03
4.	What is the difference between WHERE and HAVING clause?	2	K1	C03
5.	Write a query to get the average salary of employees by department.	2	K1	C03
6.	What does RANK() function do?	2	K1	C03
7.	How is DENSE_RANK() different from RANK()?	2	K1	C03
8.	Write a query using ROW_NUMBER() to assign a number to each employee.	2	K1	C03
9.	Write a query to find the highest salary in each department.	2	K1	C03
10	What is the use of CURRENT_DATE and CURRENT_TIME functions?	2	K1	C04
11	Give an example of using HAVING with SUM().	2	K1	C04
12	Write a query to get employee names and their ranks based on salary.	2	K1	C04
13	What is the use of DATEDIFF() in SQL? Give an example.	2	K1	C04
14	Write a query to get the minimum and maximum salary from the employees table.	2	K1	C04
15	Explain how window functions differ from aggregate functions.	2	K1	C04
PART C				
1.	Explain aggregate functions in SQL with syntax and examples.	14	K2	C04
2.	Discuss the use of GROUP BY and HAVING clauses with suitable SQL queries.	14	K2	C04
3.	Describe Window Functions in SQL. Explain RANK, DENSE_RANK, and ROW_NUMBER with examples. Explain various Date and Time functions in SQL with examples.	14	K2	C04
4.	Explain various Date and Time functions in SQL with examples.	14	K3	C04

5.	Write a detailed SQL report that shows average, min, max salary per department and ranks employees based on salary using window functions.	14	K2	CO4
----	--	----	----	-----

UNIT IV LISTS, TUPLES, DICTIONARIES				
1.	Which function is used to convert data from one type to another in SQL? a) MODIFY b) CAST c) TRANSFORM d) CHANGE	1	K1	CO5
2.	What is the purpose of COALESCE() in SQL? a) Join tables b) Remove duplicates c) Return the first non-NULL value d) Change data type	1	K1	CO5
3.	Which SQL clause is used to create a virtual table from a query? a) TEMP TABLE b) FORMAT c) VIEW d) CONVERT	1	K1	CO5
4.	What does the IS NULL condition check? a) Value is zero b) Value is NULL c) Value is empty d) Value is invalid	1	K1	CO5
5.	Which of the following is NOT a valid string function in SQL? a) CONCAT b) LENGTH c) ADDSTR d) LOWER	1	K2	CO5
6.	How do you remove spaces from both ends of a string in SQL? a) CUT() b) TRIM() c) DELETE() d) REMOVE()	1	K2	CO5
7.	Which operator is used for pattern matching in SQL? a) = b) == c) LIKE d) MATCHES	1	K2	CO5
8.	What does the % symbol represent in a LIKE clause? a) One character b) Zero or more characters c) End of string d) Digit	1	K1	CO5
9.	What function would you use to convert a string to uppercase? a) STRINGUP() b) UPPER() c) TOCAPS() d) CAPITALIZE()	1	K2	CO5
10.	Which command is used to create a temporary table in SQL? a) NEW TABLE b) CREATE TEMPORARY TABLE c) MAKE TABLE d) DEFINE TABLE	1	K1	CO5
11.	What happens when NULL is compared using = operator? a) Returns TRUE b) Returns UNKNOWN c) Returns 0 d) Returns NULL	1	K1	CO5
12.	Which function finds the position of a substring in SQL? a) LOCATE() b) POSITION() c) Both a and b d) FIND()	1	K1	CO5
13.	Which function replaces part of a string in SQL? a) CHOP() b) REPLACE() c) DELETETESTR() d) UPDATE()	1	K1	CO5
14.	Which clause is used to delete a view? a) DROP TABLE b) DROP VIEW c) DELETE VIEW	1	K2	CO5

	d) REMOVE VIEW			
15.	Which function is used to change a string to lowercase? a) DOWN() b) LOWER() c) SMALL() d) TOLOWER()	1	K2	CO5
16.	Which SQL command allows formatting numeric data? a) FORMAT() b) CAST() c) DECIMAL() d) TEXT()	1	K1	CO5
17.	What is the default value of a column if no value is provided and no default is set? a) 0 b) NULL c) Empty string d) Unknown	1	K2	CO5
18.	What is a regular expression used for? a) Pattern matching in strings b) Calculating values c) Sorting data d) Numeric conversion	1	K1	CO5
19.	Which symbol represents a single character in SQL LIKE? a) % b) _ c) * d) #	1	K1	CO5
20.	Which keyword is used to check if a value is NOT NULL? a) EXISTS b) NULL c) IS NOT NULL d) NOTNULL	1	K1	CO5
PART B				
1.	What is the use of CAST() function in SQL?	2	K1	CO5
2.	How is COALESCE() different from ISNULL() in SQL Server?	2	K1	CO5
3.	Write a query to convert a float value to integer.	2	K1	CO5
4.	How do you handle NULL values in string concatenation?	2	K1	CO5
5.	Write a query using TRIM() function.	2	K1	CO5
6.	What is the purpose of temporary tables?	2	K1	CO5
7.	Differentiate between permanent and temporary tables.	2	K1	CO5
8.	What is the syntax for creating a VIEW?	2	K1	CO5
9.	Write a query using the REPLACE() function.	2	K1	CO5
10.	What is a regular expression in SQL?	2	K1	CO5
11.	How do you check if a column has NULLs? Write an SQL query using LOWER() and UPPER() .	2	K1	CO5
12.	How do you create a temporary table for customer orders?	2	K1	CO5
13.	What does the % symbol mean in a LIKE clause?	2	K1	CO5
14.	How do you remove NULLs in SELECT output?	2	K1	CO5
15.	How do you check if a column has NULLs?	2	K1	CO5
PART C				
1.	Explain the concept of data type conversion in SQL. Give examples using CAST and CONVERT.	14	K1	CO5
2.	Discuss the handling of NULL and missing values in SQL.	14	K1	CO5
3.	Describe how views are created and managed in SQL.	14	K2	CO5

4.	Explain temporary tables in SQL. Differentiate between global and local temporary tables.	7	K1	CO5
5.	Write an SQL program that creates a view, performs string manipulation and handles NULLs using COALESCE.	7	K1	CO5
6.	Discuss various string manipulation functions in SQL with examples.	14	K1	CO5
7.	Explain regular expressions in SQL? Explain their syntax, operators, and real-world applications.	14	K1	CO5

UNIT-V FILES, MODULES, PACKAGES				
1.	Which Python module is commonly used to connect to SQLite databases? a) mysql.connector b) psycpg2 c) sqlite3 d) pandasql	1	K1	CO5
2.	Which command is used to connect to MySQL in Python? a) sqlite3.connect() b) psql.connect() c) mysql.connector.connect()** d) mysqlsql.connect()``	1	K1	CO6
3.	What is the purpose of cursor() in Python database connections? a) To close connection b) To execute SQL commands c) To view tables d) To log errors	1	K1	CO6
4.	Which of the following is a PostgreSQL connector in Python? a) sqlite3 b) psycpg2 c) pymysql d) postdb	1	K1	CO6
5.	What is %sql used for in Jupyter notebooks? a) Comment syntax b) Python magic for loops c) SQL magic command d) SQL comment	1	K1	CO6
6.	In Jupyter notebooks, which magic command allows you to run SQL code in a single line? a) %%python b) %sql c) %run d) %sqlite	1	K1	CO6
7.	What does fetchall() do in Python SQL cursor? a) Retrieves all result rows b) Executes query c) Commits transaction d) Updates the table	1	K1	CO6
8.	Which visualization tool is best known for drag-and-drop dashboard creation? a) JupyterLab b) Tableau c) MySQL Workbench d) Python IDE	1	K1	CO6
9.	Which Python library integrates SQL queries into pandas dataframes? a) SQLAlchemy b) pandasql c) scikit-learn d) dbtools	1	K1	CO6
10.	What is Power BI used for? a) Machine learning b) Data storage c) Business Intelligence dashboards d) Web scraping	1	K1	CO6
11.	Which clause in SQL helps in making visualizations effective by grouping data? a) JOIN b) GROUP BY c) ORDER d) SELECT	1	K1	CO6
12.	In Tableau, how is data imported? a) Through Python code b) Using connectors (e.g., Excel, SQL) c) Only manual input	1	K1	CO6

	d) REST API only			
13.	Which SQL function is often used for plotting sales trend over time? a) SUM() b) DROP() c) REPLACE() d) CAST()	1	K1	CO6
14.	Which feature of Power BI enables time series analysis? a) Gauge chart b) Table view c) Line chart d) Tree map	1	K1	CO6
15.	What is the extension of SQLite database files? a) .sql b) .mysql c) .db d) .dbms	1	K1	CO6
16.	Which connector would you use to connect PostgreSQL with Python? a) mysql-connector b) sqlite3 c) psycopg2 d) oracle-driver	1	K1	CO6
17.	What method is used to send a SQL command in Python? a) send() b) connect() c) execute() d) push()	1	K1	CO6
18.	In pandasql, what syntax is used to run SQL queries? a) sql.query() b) pandasql.sqldf() c) pandas.sqlrun() d) sql.run()	1	K1	CO6
19.	Which of the following is not a benefit of integrating SQL with Python? a) Automation b) Real-time analysis c) Manual data entry d) Data cleaning	1	K2	CO6
20.	Which SQL command can help analyze missing data in a data science project? a) SELECT * b) SELECT WHERE column IS NULL c) JOIN d) CAST	1	K1	CO6
PART B				
1.	What is the purpose of psycopg2 in Python?	2	K1	CO6
2.	How do you create a SQLite database using Python?	2	K2	CO6
3.	Explain the use of %sql magic command in Jupyter Notebook.	2	K1	CO6
4.	How can SQL queries be visualized using Tableau?	2	K1	CO6
5.	Write code to retrieve data from a MySQL database in Python.	2	K2	CO6
6.	What is the difference between %sql and %%sql in notebooks?	2	K1	CO6
7.	How does Power BI handle data from SQL databases?	2	K1	CO6
8.	Explain a use case of using SQL in a data science project.	2	K1	CO6
9.	What is the role of SQLAlchemy?	2	K2	CO6
10.	What is the output of fetchall() vs fetchone() ?	2	K1	CO6
11.	How do you visualize query results in Jupyter?	2	K1	CO6
12.	Write SQL to count missing values in a column.	2	K2	CO6
13.	Mention any 2 benefits of integrating SQL with Python.	2	K1	CO6
14.	How is a bar chart useful in SQL-based visualization?	2	K2	CO6
15.	Describe one method to run SQL on pandas DataFrame.	2	K1	CO6
PART C				

1.	Explain how Python can be integrated with various SQL databases like SQLite, MySQL, and PostgreSQL.	14	K3	CO6
2.	Discuss how SQL is used in Jupyter notebooks with magic commands. Include advantages and sample code.	14	K2	CO6
3.	Explain the process of visualizing SQL data using Tableau and Power BI with practical examples.	14	K2	CO6
4.	Write a complete case study: Use Python and SQL to analyze a sales dataset. Explain data extraction, cleaning, and visualization.	7	K2	CO6
5.	Compare the usage of sqlite3, mysql.connector, and psycopg2 in Python for database operations.	14	K2	CO6
6.	Demonstrate how SQL is used for data pre processing in a machine learning pipeline using Python.	14	K2	CO6
7.	Explain the role of SQL in business intelligence and data storytelling. Provide examples with Power BI/Tableau.	14	K1	CO6

Course Tutor(s)

HOD/CSE